



Scheme/ Answer Key for Valuation <i>Scheme of evaluation (marks in brackets) and answers of problems/key</i>			
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY B.TECH S1 (SPECIAL IMPROVEMENT) EXAMINATION, JANUARY 2021 (2019 SCHEME)			
Course Code: CYT100			
Course Name: ENGINEERING CHEMISTRY			
		(2019-Scheme)	
Max. Marks: 100			Duration: 3 Hours
PART A			
<i>Answer all questions, each carries 3 marks.</i>			
1		any 3 points 3 marks	(3)
2		Reactions- 1 mark , representation -1 mark emf -1 mark E^0 cell= 2.46 V	(3)
3		Requirement -1 mark 2 examples 2 marks	(3)
4		a) 2 signals 1.5 marks b) 4 signals 1.5 marks	(3)
5		In phase change no mass change, but only heat change takes place (2 marks) example 1 mark	(3)
6		Any 3 applications 3 marks	(3)
7		Ans S,-2 marks ; Fisher formula -1 mark (For fisher formula also notation must be S)	(3)
8		Definition 1 mark addition copolymer 1 marks condensation co polymer 1 mark.	(3)
9		Explanation 1 mark equations 2 marks	(3)
10		Any 3 Factors 3 marks	(3)
PART B			
<i>Answer one full question from each module, each question carries 14 marks</i>			
Module-I			
11	a)	5 types with explanation 5 marks Nernst's equation 2 marks on any one	(7)
	b)	Any 4 points -7 marks	(7)
12	a)	Principle-2 marks, explanation -3 marks . graph 2 marks	(7)
	b)	Principle-2 sacrificial anodic -3 impressed current -2	(7)
Module-II			
13	a)	NMR spectra 4 marks each, difference 2 marks	(10)
	b)	Beer Lambert's Law Equation -1mark , $A = \log I_0/I = \log 100/20 = \log 5 =$	(4)



		0.6990, $A = \epsilon c t$ $c = A / \epsilon \times t = 0.6990 / 12000 \times 0.25 \text{ dm} = 2.33 \times 10^{-4} \text{ M}$ 3 marks	
14	a)	Differentiation of NMR spectrum of $\text{CH}_3\text{CH}_2\text{Cl}$ and CH_3CHCl_2 including position-3marks, signal splitting-3 marks and intensity ratios-2marks. Signal splitting explanation using the concept spin spin splitting	(8)
	b)	2 marks each	(6)
Module-III			
15	a)	Principle-4 marks, instrumentation with diagram-3 marks, thermogram of calcium oxalate-3marks	(10)
	b)	Any four-4 marks	(4)
16	a)	Principle-2 marks, instrumentation with diagram-3 marks, procedure-3 marks, application-2 marks	(10)
	b)	Method description-3 marks, equation-1 mark	(4)
Module-IV			
17	a)	Optical isomerism – 1 mark, Conditions to be optically active – 1 mark, Definition of enantiomers – 1 mark, Example for enantiomers – 1 mark, Properties of enantiomers – 1 mark, Definition of diastereoisomers – 1 mark, Example for diastereoisomers – 1 mark, Properties of diastereoisomers – 1 mark	(8)
	b)	Structure-2 marks, synthesis-any one method-2 marks, props-1 mark, applns-1 mark	(6)
18	a)	Cis, Trans isomers-3 marks, optically active- 2 marks, conformers-3 marks, stability-2 marks	(10)
	b)	Structure of Kelvar – 2 marks, 1 mark may be given if the structures of monomers are correct. Low thermal conductivity and high strength at low temperatures – 2 marks	(4)
Module-V			
19	a)	Trickling or activated sludge with diag.-5 marks, UASB with diag.-5 marks	(10)
	b)	Temporary hardness = $(16.2 \times 100/162) + (7.3 \times 100/146) = 15 \text{ ppm}$ Permanent hardness = $(9.5 \times 100/95) + (13.6 \times 100/136) = 20 \text{ ppm}$ Total hardness = 35ppm New Temporary hardness when 10.6 mg/L NaHCO_3 is added	(4)



		$15 + (10.6 \times 100) / 2 \times 84 = 21.31 \text{ ppm}$ $\text{New permanent hardness} = 35 - 21.31 = 13.69 \text{ ppm}$	
20	a)	Principle-3 marks, procedure-4, calculation-3 marks	(10)
	b)	Any four-4 marks	(4)

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